

**NOTE:**

*A restore can be stopped at any time by pressing **[CANCEL]** or **[RESET]** except when restoring **System Data**.*

**WARNING:**

User data and memory are erased prior to a restore.

9. Press F4.

Each data area restored is checked-off and initialized.

4.5 Running Programs

Once a program is loaded on the machine and the offsets are set, to run the program:

1. Press **[CYCLE START]**.
2. It is suggested that you run the program in Graphics mode before doing any cutting.

4.6 Part Setup

Correct workholding is very important for safety, and to get the machining results that you want. There are many workholding options for different applications. Contact your HFO or workholding dealer for guidance.

4.6.1 Jog Mode

Jog Mode allows you to jog each of the axes to a desired location. Before jogging the axes it is necessary to home (beginning axes reference point) the axes.

To enter jog mode:

1. Press **[HANDLE JOG]**.
2. Pick an increment speed to be used while in jog mode (**[.0001]**, **[.001]**, **[.01]** or **[.1]**).
3. Press the desired axis (**[+X]**, **[-X]**, **[+Z]**, or **[-Z]**) and either press and hold these axis jog keys or use the **[HANDLE JOG]** control to move the selected axis.

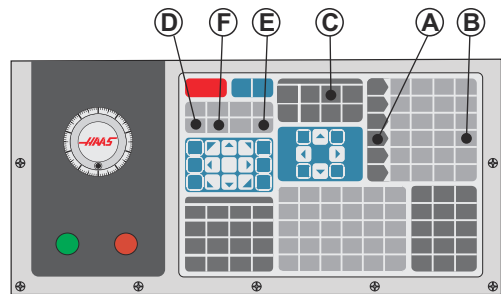
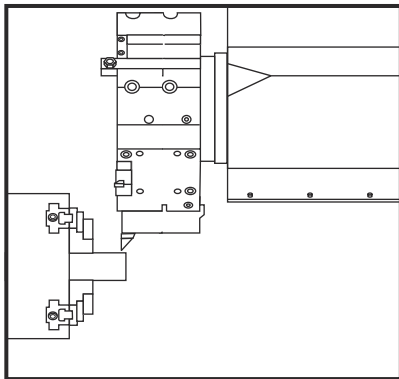
4.6.2 Setting the Tool Offset

The next step is to touch off the tools. Doing this defines the distance from the tip of the tool to the side of the part. This procedure requires the following:

- An O.D. Turning Tool
- A work piece that fits in the chuck jaws
- A measuring tool to inspect the workpiece diameter

For information on setting up Live tools, refer to page 193.

F4.10: Lathe Tool Offset



1. Press **[OFFSET]**. Press **[HANDLE JOG]**.
2. Load an O.D. turning tool into the tool turret. Press **[NEXT TOOL]** [F] until it is the current tool.
3. Clamp the workpiece in the spindle.
4. Press **[.1/100]** [B]. The axis selected moves at a fast rate when the handle is turned.
5. Close the lathe door. Type 50 and press **[FWD]** for the spindle to start.
6. Use the turning tool loaded in station 1 to make a small cut on the diameter of the material clamped in the spindle. Approach the part carefully and feed slowly during the cut.
7. After the small cut is done, jog away from the part using Z-Axis. Move far enough away from the part so that you can take a measurement with your measuring tool.
8. Press Spindle **[STOP]** and open the door.
9. Use the measuring tool to measure the cut made on the workpiece
10. Press **[X DIAMETER MEASURE]** [D] to record the X-axis position in the offset table.
11. Type the workpiece diameter and press **[ENTER]** to add it to the X-axis offset. The offset that corresponds to the tool and turret station is recorded.
12. Close the lathe door. Type 50 and press **[FWD]** for the spindle to start.
13. Use the turning tool loaded in station 1 to make a small cut on the face of the material clamped in the spindle. Approach the part carefully and feed slowly during the cut.
14. After the small cut is done, jog away from the part using X-axis. Move far enough away from the part so that you can take a measurement with your measuring tool.
15. Press **[Z FACE MEASURE]** (E) to record the current Z position in the offset table.
16. The cursor moves to the Z-axis location for the tool.
17. Repeat all of the previous steps for each tool in the program. Do tool changes at a safe location with no obstructions.